

Problem 2: GCD-Sidon Sets

A Resolution of the Square-Root Question and a Refined Follow-Up Problem

Draft note for discussion

23 May 2026

Abstract

We consider the GCD-Sidon problem from the note *Ten New Erdős-style problems in number theory and combinatorics*. The original question asked whether the largest GCD-Sidon subset of $[N]$ has size $N^{1/2+o(1)}$, and whether its ratio to $\sqrt{N}$ has a limit. We give a short local divisor argument proving the much stronger bound

$$G(N) \leq 1 + \max_{n \leq N} \tau(n) = N^{o(1)}.$$

Therefore the proposed square-root asymptotic is false, and in fact $G(N)/\sqrt{N} \rightarrow 0$. We also give a lower bound of the form

$$G(N) \geq \exp\left(\left(\frac{\log 2}{2} + o(1)\right) \frac{\log N}{\log \log N}\right),$$

which motivates the refined problem of determining

$$c_- := \liminf_{N \rightarrow \infty} \frac{\log G(N)}{\log N / \log \log N}, \quad c_+ := \limsup_{N \rightarrow \infty} \frac{\log G(N)}{\log N / \log \log N}.$$

The arguments here give

$$\frac{\log 2}{2} \leq c_- \leq c_+ \leq \log 2.$$

1 The original problem

For $N \geq 1$, write

$$[N] = \{1, 2, \dots, N\}.$$

For positive integers a, b , write (a, b) for their greatest common divisor. Let $\tau(n)$ denote the number of positive divisors of n .

Problem 1 (GCD-Sidon sets). *Call $A \subseteq [N]$ GCD-Sidon if all values*

$$(a, b), \quad a < b, \quad a, b \in A,$$

are distinct. Let

$$G(N) = \max\{|A| : A \subseteq [N] \text{ is GCD-Sidon}\}.$$

The original questions are:

(a) *Is*

$$G(N) = N^{1/2+o(1)}?$$

(b) *Does*

$$\frac{G(N)}{\sqrt{N}}$$

have a limit?

The elementary upper bound in the original problem statement is

$$\binom{|A|}{2} \leq N,$$

since the $\binom{|A|}{2}$ pairwise gcds are distinct elements of $[N]$. Hence

$$G(N) \leq \sqrt{2N} + 1.$$

This counting bound suggests a possible square-root barrier. The key observation below is that there is a much stronger *local* obstruction: after fixing one element $a \in A$, all gcds involving a must be distinct divisors of a .

2 Resolution of the square-root question

Define

$$D(N) = \max_{1 \leq n \leq N} \tau(n).$$

Theorem 1 (Local divisor upper bound). *For every $N \geq 1$,*

$$G(N) \leq 1 + D(N).$$

Consequently,

$$G(N) = N^{o(1)}.$$

In particular,

$$\frac{G(N)}{\sqrt{N}} \rightarrow 0.$$

Thus the proposed estimate $G(N) = N^{1/2+o(1)}$ is false, and the original ratio question has the limiting answer 0.

Proof. Let $A \subseteq [N]$ be GCD-Sidon, and write

$$|A| = m.$$

If $m \leq 1$, there is nothing to prove. Assume $m \geq 2$ and fix one element

$$a \in A.$$

For every other element $b \in A \setminus \{a\}$, define

$$d_b = (a, b).$$

Each d_b is a positive divisor of a , so

$$d_b \in \text{Div}(a) := \{d \in \mathbb{N} : d \mid a\}.$$

We claim that the map

$$A \setminus \{a\} \longrightarrow \text{Div}(a), \quad b \longmapsto d_b = (a, b),$$

is injective. Indeed, suppose that $b, c \in A \setminus \{a\}$ are distinct and that

$$d_b = d_c.$$

Then

$$(a, b) = (a, c).$$

The unordered pairs $\{a, b\}$ and $\{a, c\}$ are distinct, because $b \neq c$. Thus two distinct unordered pairs of elements of A would have the same gcd. This contradicts the GCD-Sidon property.

Therefore $A \setminus \{a\}$ injects into the divisor set of a . Hence

$$m - 1 \leq |\text{Div}(a)| = \tau(a).$$

Since $a \leq N$, we also have

$$\tau(a) \leq D(N).$$

Thus

$$m \leq 1 + D(N).$$

Taking the maximum over all GCD-Sidon sets $A \subseteq [N]$ gives

$$G(N) \leq 1 + D(N).$$

It remains to use the standard divisor estimate

$$D(N) = N^{o(1)}.$$

For completeness, Lemma 1 gives a self-contained proof of the weaker form $\tau(n) = n^{o(1)}$, which is enough here. Applying that lemma uniformly for $n \leq N$ gives

$$D(N) = N^{o(1)}.$$

Therefore

$$G(N) \leq 1 + D(N) = N^{o(1)}.$$

Since $G(N) \geq 1$, this proves $G(N) = N^{o(1)}$ in the sense that $\log G(N)/\log N \rightarrow 0$.

Finally,

$$0 \leq \frac{G(N)}{\sqrt{N}} \leq \frac{1 + D(N)}{\sqrt{N}} = N^{-1/2+o(1)} \longrightarrow 0.$$

So $G(N)/\sqrt{N}$ has limit 0. □

Lemma 1 (Elementary divisor bound). *For every fixed $\varepsilon > 0$ there is $n_0(\varepsilon)$ such that, for every $n \geq n_0(\varepsilon)$,*

$$\tau(n) \leq n^\varepsilon.$$

Equivalently, $\tau(n) = n^{o(1)}$.

Proof. Fix $\varepsilon > 0$. Choose $P = P(\varepsilon)$ so large that

$$2 \leq p^{\varepsilon/2}$$

for every prime $p > P$.

Write

$$n = \prod_p p^{\alpha_p}$$

for the prime factorisation of n . Then

$$\tau(n) = \prod_p (\alpha_p + 1).$$

We split this product into primes $p \leq P$ and primes $p > P$.

If $p > P$ and $\alpha_p \geq 1$, then

$$\alpha_p + 1 \leq 2^{\alpha_p} \leq p^{(\varepsilon/2)\alpha_p}.$$

Therefore the contribution from the large primes is at most

$$\prod_{p>P} (\alpha_p + 1) \leq \prod_{p>P} p^{(\varepsilon/2)\alpha_p} \leq n^{\varepsilon/2}.$$

For the finitely many primes $p \leq P$, we use the crude bound

$$\alpha_p \leq \frac{\log n}{\log 2}.$$

Hence

$$\prod_{p \leq P} (\alpha_p + 1) \leq \left(\frac{\log n}{\log 2} + 1 \right)^{\pi(P)}.$$

Since P is fixed once ε is fixed, the right-hand side is a fixed power of $\log n$. For all sufficiently large n , this fixed power is at most $n^{\varepsilon/2}$. Combining the two estimates gives

$$\tau(n) = \prod_{p \leq P} (\alpha_p + 1) \prod_{p > P} (\alpha_p + 1) \leq n^{\varepsilon/2} n^{\varepsilon/2} = n^{\varepsilon}$$

for all sufficiently large n . □

3 The sharper divisor-function scale

Theorem 1 already resolves the original square-root question. To identify the more natural scale, recall Wigert's classical maximal-order theorem for the divisor function:

$$D(N) = \max_{n \leq N} \tau(n) = \exp\left((\log 2 + o(1)) \frac{\log N}{\log \log N}\right).$$

Together with Theorem 1, this yields

$$G(N) \leq \exp\left((\log 2 + o(1)) \frac{\log N}{\log \log N}\right). \tag{1}$$

The next construction shows that this scale is not an artifact of the upper bound: $G(N)$ is also at least exponential in $\log N / \log \log N$.

Theorem 2 (A binary Sidon-code lower bound). *As $N \rightarrow \infty$,*

$$G(N) \geq \exp\left(\left(\frac{\log 2}{2} + o(1)\right) \frac{\log N}{\log \log N}\right).$$

Proof. The proof has three steps. First we build a binary Sidon code. Then we encode its coordinates by pairs of primes. Finally we choose the number of coordinates so that all constructed integers lie in $[N]$.

Step 1: a binary Sidon code. Let $s \geq 1$, and let $K = \mathbb{F}_{2^s}$ be the finite field of order 2^s . Consider

$$C_0 = \{(x, x^3) : x \in K\} \subseteq K^2.$$

We claim that C_0 is a Sidon set in the additive group of K^2 : if

$$(x, x^3) + (y, y^3) = (u, u^3) + (v, v^3)$$

with $x \neq y$ and $u \neq v$, then

$$\{x, y\} = \{u, v\}.$$

Indeed, equality of first coordinates gives

$$S := x + y = u + v.$$

Since $x \neq y$ and the characteristic is 2, we have $S \neq 0$. Equality of second coordinates gives

$$T := x^3 + y^3 = u^3 + v^3.$$

In characteristic 2,

$$x^3 + y^3 = (x + y)(x^2 + xy + y^2).$$

Also

$$x^2 + y^2 = (x + y)^2 = S^2.$$

Therefore

$$T = S(S^2 + xy).$$

Since $S \neq 0$, the product xy is determined by S and T :

$$xy = \frac{T}{S} + S^2.$$

Thus x and y are the two roots of

$$Z^2 + SZ + xy = 0,$$

so the unordered pair $\{x, y\}$ is determined by the group sum. The same applies to u, v , proving $\{x, y\} = \{u, v\}$.

Choose an $\mathbb{F}_2$ -basis of K and identify K^2 with $\mathbb{F}_2^{2s} = \{0, 1\}^{2s}$. Let $C \subseteq \{0, 1\}^{2s}$ be the binary image of C_0 . Then

$$|C| = 2^s.$$

Moreover, if two unordered pairs of distinct codewords satisfy

$$c + d = e + f$$

as an equality of coordinatewise ordinary integer sums in $\{0, 1, 2\}^{2s}$, then reducing modulo 2 gives the same equality in $\mathbb{F}_2^{2s}$. By the Sidon property just proved,

$$\{c, d\} = \{e, f\}.$$

Thus the coordinatewise integer sums $c + d$ distinguish unordered pairs of distinct codewords.

Step 2: prime-coordinate encoding. Put

$$r = 2s.$$

Choose $2r$ distinct primes, arranged as ordered pairs

$$(p_{1,0}, p_{1,1}), \dots, (p_{r,0}, p_{r,1}).$$

For each codeword $c = (c_1, \dots, c_r) \in C$, define

$$a_c = \prod_{i=1}^r p_{i,c_i}.$$

Let

$$A_C = \{a_c : c \in C\}.$$

By unique factorisation, $c \mapsto a_c$ is injective, so

$$|A_C| = |C| = 2^s = 2^{r/2}.$$

We claim that A_C is GCD-Sidon. Suppose that $\{c, d\}$ and $\{e, f\}$ are unordered pairs of distinct codewords and that

$$(a_c, a_d) = (a_e, a_f).$$

Fix a coordinate i . In the gcd (a_c, a_d) :

- the prime $p_{i,0}$ occurs if and only if $c_i = d_i = 0$;
- the prime $p_{i,1}$ occurs if and only if $c_i = d_i = 1$;
- neither $p_{i,0}$ nor $p_{i,1}$ occurs if and only if $c_i \neq d_i$.

Since all primes $p_{i,j}$ are distinct, equality of the two gcds forces the same alternative for (e_i, f_i) . Therefore, for every coordinate i ,

$$c_i + d_i = e_i + f_i$$

as an equality in $\{0, 1, 2\}$. Hence

$$c + d = e + f$$

as coordinatewise integer sums. By the binary Sidon property of C ,

$$\{c, d\} = \{e, f\}.$$

Thus the unordered pairs $\{a_c, a_d\}$ and $\{a_e, a_f\}$ are equal. Hence distinct unordered pairs of elements of A_C have distinct gcds, so A_C is GCD-Sidon.

Step 3: keeping the construction inside $[N]$. Let q_j be the j th prime, and choose, for example,

$$p_{i,0} = q_i, \quad p_{i,1} = q_{r+i} \quad (1 \leq i \leq r).$$

Then every a_c is at most

$$\prod_{i=1}^r q_{r+i} = \prod_{j=r+1}^{2r} q_j.$$

By the prime number theorem,

$$\sum_{j=r+1}^{2r} \log q_j = (1 + o(1))r \log r.$$

Therefore

$$\prod_{j=r+1}^{2r} q_j = \exp((1 + o(1))r \log r).$$

Choose an even integer r with

$$r = (1 + o(1)) \frac{\log N}{\log \log N}$$

and

$$\prod_{j=r+1}^{2r} q_j \leq N.$$

For instance, one may take

$$r = 2 \left\lfloor \frac{1}{2} \cdot \frac{\log N}{\log \log N + 2 \log \log \log N} \right\rfloor$$

for all sufficiently large N . With this choice, $A_C \subseteq [N]$. Its size is

$$|A_C| = 2^{r/2},$$

and hence

$$\log |A_C| = \frac{r}{2} \log 2 = \left(\frac{\log 2}{2} + o(1) \right) \frac{\log N}{\log \log N}.$$

This proves the lower bound. □

Combining Theorem 2 with the upper bound (1), we obtain

$$\exp\left(\left(\frac{\log 2}{2} + o(1)\right) \frac{\log N}{\log \log N}\right) \leq G(N) \leq \exp\left((\log 2 + o(1)) \frac{\log N}{\log \log N}\right). \quad (2)$$

4 A sensible follow-up problem

The original normalization $G(N)/\sqrt{N}$ is now too coarse: it always tends to 0. The natural replacement is to ask for the precise constant on the scale visible in (2).

Define

$$\gamma(N) = \frac{\log G(N)}{\log N / \log \log N}.$$

Then define

$$c_- := \liminf_{N \rightarrow \infty} \gamma(N), \quad c_+ := \limsup_{N \rightarrow \infty} \gamma(N).$$

Problem 2 (Refined GCD-Sidon constant problem). *Determine c_- and c_+ . In particular, decide whether*

$$c_- = c_+.$$

Equivalently, decide whether there exists a constant c such that

$$G(N) = \exp\left((c + o(1)) \frac{\log N}{\log \log N}\right).$$

If such a constant exists, determine it.

The bounds proved above give

$$\frac{\log 2}{2} \leq c_- \leq c_+ \leq \log 2.$$

The upper bound is inherited from the maximal order of the divisor function. The lower bound comes from binary Sidon codes encoded by prime pairs.

Remark 1 (Why this is the right follow-up). The proof of Theorem 1 shows that the original square-root scale is not the right scale: one fixed element $a \in A$ already limits the rest of A to at most $\tau(a)$ possible gcd values. Thus the problem naturally lives on the maximal-divisor scale

$$\exp\left(\Theta\left(\frac{\log N}{\log \log N}\right)\right).$$

The lower-bound construction suggests a structural interpretation. If an integer is represented by its prime-exponent vector

$$n = \prod_i p_i^{x_i},$$

then

$$\gcd(n, m) = \prod_i p_i^{\min(x_i, y_i)}.$$

Thus GCD-Sidon sets are weighted *meet-Sidon* families in a product of chains: all pairwise coordinatewise meets must be distinct. The refined constants c_- and c_+ measure the asymptotic capacity of such weighted meet-Sidon families under the weight constraint

$$\sum_i x_i \log p_i \leq \log N.$$

Determining whether $c_- = c_+$, and computing the value if they coincide, is the natural nontrivial problem left after the original square-root question is resolved.

5 Summary

The original GCD-Sidon problem asked whether $G(N)$ has square-root order and whether $G(N)/\sqrt{N}$ has a limit. The answers are:

(a) No: in fact

$$G(N) = N^{o(1)}.$$

(b) Yes: in fact

$$\frac{G(N)}{\sqrt{N}} \rightarrow 0.$$

The proposed follow-up problem is to determine

$$c_- = \liminf_{N \rightarrow \infty} \frac{\log G(N)}{\log N / \log \log N}, \quad c_+ = \limsup_{N \rightarrow \infty} \frac{\log G(N)}{\log N / \log \log N}.$$

At present the elementary bounds in this note give

$$\frac{\log 2}{2} \leq c_- \leq c_+ \leq \log 2.$$

References

- [1] P. Chojecki and GPT-5.5 Pro, *Ten New Erdős-style problems in number theory and combinatorics*, May 2026.
- [2] S. Wigert, *Sur l'ordre de grandeur du nombre des diviseurs d'un entier*, Arkiv för Matematik, Astronomi och Fysik **3** (1907), no. 18, 1–9.
- [3] G. H. Hardy and E. M. Wright, *An Introduction to the Theory of Numbers*, 6th ed., Oxford University Press, 2008.